

Frozen-History Matrix Fields for Stochastic Nesterov Muon: An ODE Approximation

Abstract

Muon combines a matrix-valued momentum state with a nonlinear orthogonalization map. For fixed momentum parameter and vanishing outer step size, these variables evolve on different time scales: the momentum mixes in discrete time while the parameter matrix moves slowly. We freeze the parameter, run the momentum from the infinite past, and average the Nesterov-corrected orthogonalized direction. The resulting deterministic mapping is called the *Nesterov Muon matrix field*. For bounded Lipschitz stochastic gradients and any bounded Lipschitz orthogonalization map, we prove that the piecewise-linear interpolation of stochastic Nesterov Muon tracks the field ODE in L^2 , uniformly on compact time intervals, with error $O(\sqrt{\eta})$. Stabilized finite-step Newton–Schulz maps used by practical Muon satisfy the required regularity. The proof is based on blockwise freezing, geometric forgetting of momentum, a bounded difference estimate for frozen blocks, and a Hilbert-space martingale maximal inequality. Finally, for a quadratic objective with additive isotropic Gaussian matrix noise and exact polar orthogonalization, we identify the matrix field as the gradient of a Gaussian-smoothed nuclear norm. This gives an explicit Lyapunov function and shows precisely how the Nesterov correction changes the smoothing scale.

1 Introduction

Muon orthogonalizes a matrix-valued momentum by a small number of Newton–Schulz iterations before applying the parameter update [1, 2]. The method has been used at language-model scale [3], and its geometry can be interpreted through steepest descent in the spectral norm and nuclear–spectral duality [4, 5]. Recent analyses establish stationarity bounds for exact or inexact polar updates, including finite-step Newton–Schulz and Nesterov momentum [6, 7]. Those results answer finite-horizon optimization questions. The present paper asks a different dynamical question:

For fixed momentum and vanishing outer step size, which deterministic matrix field describes the macroscopic motion of stochastic Nesterov Muon?

This question is motivated by the Adam vector field of Dereich and Jentzen [8] and the corresponding fast–slow ODE approximation [9]. The relevant drift is not obtained by applying an orthogonalization map to the mean gradient. Momentum retains a random infinite history, the current sample appears both in the updated momentum and in the Nesterov correction, and orthogonalization is nonlinear. The correct order is therefore to freeze the parameter, form stationary momentum, orthogonalize the Nesterov input, and only then average.

Our contributions are the following.

- (i) We define the frozen-history Nesterov Muon matrix field, retaining the dependence between the current stochastic gradient and the current term in momentum.
- (ii) We prove boundedness and Lipschitz continuity of the field and verify that every fixed-step stabilized Newton–Schulz map is a valid orthogonalization map for the theory.

- (iii) We prove a quantitative finite-time homogenization theorem. If W^η denotes the stochastic interpolation and w solves the field ODE, then for every $T < \infty$,

$$\left(\mathbb{E} \left[\sup_{0 \leq t \leq T} \|W^\eta(t) - w(t)\|_F^2 \right] \right)^{1/2} \leq C_T \sqrt{\eta}.$$

- (iv) In the additive Gaussian quadratic model, we show that the expected exact-polar field is the gradient of a Gaussian convolution of the nuclear norm. The Nesterov correction changes only the convolution variance, which we compute in closed form.

The main theorem treats fixed $\beta < 1$. The scaling $\beta = 1 - \gamma\eta$ leads instead to a coupled inertial limit and is not covered here. Likewise, the exact polar map is singular at rank-deficient matrices and is not used in the tracking theorem. It is treated separately in the Gaussian model, where averaging regularizes the field.

2 Algorithm and orthogonalization maps

Let $\mathbb{H} = \mathbb{R}^{m \times n}$ with Frobenius inner product and norm. Let $(U_k)_{k \geq 1}$ be i.i.d. innovations and let $G : \mathcal{U} \times \mathbb{H} \rightarrow \mathbb{H}$ be a measurable stochastic gradient. For $\beta \in [0, 1)$ and $\eta > 0$, normalized Nesterov Muon is

$$M_{k+1} = \beta M_k + (1 - \beta)G(U_{k+1}, W_k), \quad (1)$$

$$N_{k+1} = \beta M_{k+1} + (1 - \beta)G(U_{k+1}, W_k), \quad (2)$$

$$W_{k+1} = W_k - \eta \mathcal{Q}(N_{k+1}). \quad (3)$$

In particular,

$$N_{k+1} = \beta^2 M_k + (1 - \beta^2)G(U_{k+1}, W_k). \quad (4)$$

The two linear interpolations in the reference implementation [2] are exactly (1)–(2). The unnormalized convention $C_{k+1} = \beta C_k + G_{k+1}$ and $\beta C_{k+1} + G_{k+1}$ differs by the common factor $1 - \beta$. The fixed layer-shape multiplier used by the reference implementation can be absorbed into $\mathcal{Q}$ throughout.

2.1 Exact and finite-step maps

If $X = U\Sigma V^\top$ is a compact singular value decomposition, its polar factor is

$$\text{Polar}(X) = UV^\top.$$

It solves

$$\text{Polar}(X) \in \operatorname{argmax}_{\|D\|_{\text{op}} \leq 1} \langle X, D \rangle_F, \quad \langle X, \text{Polar}(X) \rangle_F = \|X\|_*.$$

The map is not continuous at matrices where the polar factor is nonunique.

Practical Muon first normalizes and then applies a finite matrix polynomial. To make the regularity needed below explicit, define

$$S_\varepsilon(X) = \frac{X}{\|X\|_F + \varepsilon}, \quad \varepsilon > 0, \quad (5)$$

and, after transposing when necessary so that the number of rows does not exceed the number of columns,

$$P(X) = aX + b(XX^\top)X + c(XX^\top)^2X. \quad (6)$$

The stabilized q -step map is

$$\mathcal{Q}_{q,\varepsilon} = P^{\circ q} \circ S_\varepsilon. \quad (7)$$

The practical quintic coefficients are $(a, b, c) = (3.4445, -4.7750, 2.0315)$; classical convergent choices include the cubic and quintic Newton–Schulz polynomials discussed in [11, 6, 7].

Proposition 2.1 (Finite-step Newton–Schulz is bounded and Lipschitz). *For fixed $q < \infty$ and $\varepsilon > 0$, the map $\mathcal{Q}_{q,\varepsilon} : \mathbb{H} \rightarrow \mathbb{H}$ is globally bounded and globally Lipschitz.*

Proof. The normalization satisfies $\|S_\varepsilon(X)\|_F \leq 1$ and

$$\|S_\varepsilon(X) - S_\varepsilon(Y)\|_F \leq \frac{2}{\varepsilon} \|X - Y\|_F.$$

Set $B_0 = 1$ and

$$B_{j+1} = |a|B_j + |b|B_j^3 + |c|B_j^5.$$

Then every j th iterate is contained in the Frobenius ball of radius B_j . On the ball of radius B , telescoping the three and five matrix factors gives

$$\|P(X) - P(Y)\|_F \leq (|a| + 3|b|B^2 + 5|c|B^4) \|X - Y\|_F.$$

Composition proves the claim, with output bound B_q and a finite Lipschitz constant bounded by

$$\frac{2}{\varepsilon} \prod_{j=0}^{q-1} (|a| + 3|b|B_j^2 + 5|c|B_j^4).$$

□

3 The frozen-history matrix field

Let $(U_j)_{j \in \mathbb{Z}}$ be a two-sided i.i.d. sequence. At a fixed $W \in \mathbb{H}$ define the stationary frozen momentum

$$\mathbf{M}_\beta(W) = (1 - \beta) \sum_{j=0}^{\infty} \beta^j G(U_{-j}, W), \quad (8)$$

and the frozen Nesterov input

$$\mathbf{N}_\beta(W) = \beta \mathbf{M}_\beta(W) + (1 - \beta) G(U_0, W). \quad (9)$$

The same U_0 occurs in both terms. Expanding the series gives

$$\mathbf{N}_\beta(W) = (1 - \beta^2) G(U_0, W) + (1 - \beta) \sum_{j=1}^{\infty} \beta^{j+1} G(U_{-j}, W). \quad (10)$$

Definition 3.1 (Nesterov Muon matrix field). *Whenever the expectation exists, define*

$$\mathcal{F}_\beta(W) = \mathbb{E}[\mathcal{Q}(\mathbf{N}_\beta(W))]. \quad (11)$$

The associated field ODE is

$$\dot{w}(t) = -\mathcal{F}_\beta(w(t)), \quad w(0) = w_0. \quad (12)$$

It is useful to write the same object using the frozen momentum Markov chain. For $w, m \in \mathbb{H}$ and $u \in \mathbb{U}$, let

$$T_w(m, u) = \beta m + (1 - \beta)G(u, w), \quad (13)$$

$$H_w(m, u) = \mathcal{Q}(\beta^2 m + (1 - \beta^2)G(u, w)). \quad (14)$$

The chain $m_{j+1} = T_w(m_j, U_{j+1})$ has stationary version

$$m_j^w = (1 - \beta) \sum_{\ell=0}^{\infty} \beta^\ell G(U_{j-\ell}, w),$$

and

$$\mathcal{F}_\beta(w) = \mathbb{E}[H_w(m_{-1}^w, U_0)]. \quad (15)$$

Assumption 3.2 (Bounded Lipschitz data). *There are constants $G_\star, L_G, B_Q, L_Q < \infty$ such that for all u, w, v, x, y ,*

$$\begin{aligned} \|G(u, w)\|_F &\leq G_\star, & \|G(u, w) - G(u, v)\|_F &\leq L_G \|w - v\|_F, \\ \|\mathcal{Q}(x)\|_F &\leq B_Q, & \|\mathcal{Q}(x) - \mathcal{Q}(y)\|_F &\leq L_Q \|x - y\|_F. \end{aligned}$$

Proposition 3.3 (Regularity of the matrix field). *Under Assumption 3.2, the series (8) converges absolutely almost surely and in every L^p , $p < \infty$. Moreover,*

$$\|\mathcal{F}_\beta(w)\|_F \leq B_Q, \quad \|\mathcal{F}_\beta(w) - \mathcal{F}_\beta(v)\|_F \leq L_Q L_G \|w - v\|_F.$$

Consequently, (12) has a unique global solution.

Proof. Absolute convergence follows from the geometric weights and the bound on G . Couple the frozen histories at w and v using the same innovations. Since the weights in (8) sum to one,

$$\|\mathbf{M}_\beta(w) - \mathbf{M}_\beta(v)\|_F \leq L_G \|w - v\|_F.$$

Equation (9) gives the same bound for $\mathbf{N}_\beta$. Apply the Lipschitz property of $\mathcal{Q}$, then Jensen's inequality. Global existence follows from the global Lipschitz and boundedness estimates. $\square$

Proposition 3.4 (Deterministic collapse). *If $G(U, w) = g(w)$ almost surely, then*

$$\mathbf{M}_\beta(w) = \mathbf{N}_\beta(w) = g(w), \quad \mathcal{F}_\beta(w) = \mathcal{Q}(g(w)).$$

Thus the distinction between ordinary and Nesterov momentum in the frozen field is entirely stochastic.

Proof. Both sets of geometric coefficients sum to one. $\square$

4 Finite-time ODE approximation

We now prove the main result. Let $\mathcal{G}_k = \sigma(W_0, M_0, U_1, \dots, U_k)$. Assume $W_0 = w_0$ is deterministic, M_0 is $\mathcal{G}_0$ -measurable, and

$$\|M_0\|_F \leq K_M \quad \text{almost surely.} \quad (16)$$

Let W^η be the piecewise-linear interpolation satisfying $W^\eta(k\eta) = W_k$.

Theorem 4.1 (Quantitative tracking of the matrix-field ODE). *Suppose Assumption 3.2 and (16) hold, and let w solve (12). For every $T < \infty$ there is a finite constant*

$$C_T = C(T, \beta, G_\star, L_G, B_Q, L_Q, K_M)$$

such that, for every $\eta \in (0, 1]$,

$$\left(\mathbb{E} \left[\sup_{0 \leq t \leq T} \|W^\eta(t) - w(t)\|_F^2 \right] \right)^{1/2} \leq C_T \sqrt{\eta}. \quad (17)$$

In particular, $W^\eta \rightarrow w$ uniformly on $[0, T]$ in L^2 and in probability.

The proof uses the next frozen-block estimate. It is stated conditionally in a form that can be applied at a random block anchor.

Lemma 4.2 (Bias and fluctuations in a frozen block). *Fix $w \in \mathbb{H}$ and $m \in \mathbb{H}$ with $\|m\|_F \leq K$, and let*

$$m_j = T_w(m_{j-1}, U_j), \quad Z_j = H_w(m_{j-1}, U_j), \quad m_0 = m.$$

Then for every $h \geq 1$,

$$\left\| \mathbb{E} \left[\sum_{j=1}^h (Z_j - \mathcal{F}_\beta(w)) \right] \right\|_F \leq C_{\text{mix}}, \quad (18)$$

$$\mathbb{E} \left[\left\| \sum_{j=1}^h Z_j - \mathbb{E} \left[\sum_{j=1}^h Z_j \right] \right\|_F^2 \right] \leq C_{\text{var}} h, \quad (19)$$

where one may take

$$C_{\text{mix}} = \frac{L_Q(K + G_\star)\beta^2}{1 - \beta}, \quad C_{\text{var}} = 4L_Q^2 G_\star^2.$$

Proof. Couple the chain started at m with a stationary frozen chain, using the same future innovations. Their pre-update states at time j differ by $\beta^{j-1}(m - m_0^w)$. Since $\|m_0^w\|_F \leq G_\star$ and H_w has Lipschitz constant $L_Q\beta^2$ in its momentum argument,

$$\|\mathbb{E}Z_j - \mathcal{F}_\beta(w)\|_F \leq L_Q\beta^{j+1}(K + G_\star).$$

Summing the geometric series proves (18).

For (19), expose $U_1, \dots, U_h$ one at a time and use the Doob martingale for $\sum_{j=1}^h Z_j$. If only U_ℓ is replaced by an independent copy, the change at time ℓ is at most $2L_Q G_\star(1 - \beta^2)$. Its effect on the momentum before time $j > \ell$ is at most $2G_\star(1 - \beta)\beta^{j-1-\ell}$, so the sum of all future effects is at most

$$2L_Q G_\star(1 - \beta) \sum_{r=1}^{\infty} \beta^{r+1} = 2L_Q G_\star \beta^2.$$

Thus each Doob difference has norm at most $2L_Q G_\star$. Orthogonality of Hilbert-space martingale differences yields (19). $\square$

Proof of Theorem 4.1. Write $D_{k+1} = \mathcal{Q}(N_{k+1})$. Since $\|D_k\|_F \leq B_Q$,

$$\|W_k - W_\ell\|_F \leq \eta B_Q |k - \ell|. \quad (20)$$

Fix an integer block length h and anchors $a_r = rh$. On the block $a_r \leq k < a_{r+1}$, define a frozen chain driven by the same innovations,

$$\begin{aligned}\widehat{M}_{a_r} &= M_{a_r}, \\ \widehat{M}_{k+1} &= \beta \widehat{M}_k + (1 - \beta)G(U_{k+1}, W_{a_r}), \\ \widehat{D}_{k+1} &= H_{W_{a_r}}(\widehat{M}_k, U_{k+1}).\end{aligned}$$

Equations (20) and (1) imply

$$\|M_k - \widehat{M}_k\|_F \leq L_G \eta B_Q h.$$

Using (4),

$$\|D_{k+1} - \widehat{D}_{k+1}\|_F \leq L_Q L_G \eta B_Q h. \quad (21)$$

Proposition 3.3 also gives

$$\|\mathcal{F}_\beta(W_k) - \mathcal{F}_\beta(W_{a_r})\|_F \leq L_Q L_G \eta B_Q h. \quad (22)$$

It remains to control the centered frozen directions. Conditional on $\mathcal{G}_{a_r}$, Lemma 4.2 applies with $K = \max\{K_M, G_\star\}$. For a complete block set

$$S_r = \sum_{k=a_r}^{a_{r+1}-1} (\widehat{D}_{k+1} - \mathcal{F}_\beta(W_{a_r})), \quad \Delta_r = S_r - \mathbb{E}[S_r \mid \mathcal{G}_{a_r}].$$

Then $(\Delta_r)_r$ is a martingale-difference sequence at block boundaries,

$$\|\mathbb{E}[S_r \mid \mathcal{G}_{a_r}]\|_F \leq C_{\text{mix}}, \quad \mathbb{E}[\|\Delta_r\|_F^2 \mid \mathcal{G}_{a_r}] \leq C_{\text{var}} h.$$

If R complete blocks lie before time T , the Hilbert-space Doob inequality gives

$$\begin{aligned}\left(\mathbb{E} \left[\max_{0 \leq s \leq R} \left\| \eta \sum_{r=0}^{s-1} S_r \right\|_F^2 \right] \right)^{1/2} &\leq \eta R C_{\text{mix}} + 2\eta \sqrt{R C_{\text{var}} h} \\ &\leq C_T (h^{-1} + \sqrt{\eta}).\end{aligned} \quad (23)$$

An incomplete block contributes at most $2B_Q \eta h$.

Combining (21), (22), and (23), we obtain a remainder R_k^η such that, for $k\eta \leq T$,

$$W_k = W_0 - \eta \sum_{j=0}^{k-1} \mathcal{F}_\beta(W_j) + R_k^\eta, \quad \left(\mathbb{E} \max_{k\eta \leq T} \|R_k^\eta\|_F^2 \right)^{1/2} \leq C_T (\sqrt{\eta} + h^{-1} + \eta h). \quad (24)$$

Choose $h = \lceil \eta^{-1/2} \rceil$. A discrete Gronwall inequality compares (24) with the Euler scheme for (12), multiplying the right-hand side by at most $\exp(L_Q L_G T)$. The deterministic Euler error and the linear-interpolation error are $O(\eta)$ because the field is bounded and Lipschitz. This proves (17). $\square$

Remark 4.3 (Localization). *Global boundedness is used to keep the principal theorem transparent. If the gradient and orthogonalization map are only locally bounded and locally Lipschitz, the same proof applies up to the first exit from a fixed compact set. A global result then follows from a separate tightness or stability estimate for the iterates.*

Remark 4.4 (Weight decay). *For the decoupled update $W_{k+1} = (1 - \eta\lambda)W_k - \eta\mathcal{Q}(N_{k+1})$, the limit is $\dot{w} = -\mathcal{F}_\beta(w) - \lambda w$. On finite time intervals the proof changes only through the Lipschitz constant and the elementary bound replacing (20).*

5 Quadratic Gaussian model and a smoothed nuclear-norm potential

The main theorem deliberately avoids the singular exact polar map. We now show that nondegenerate noise can regularize the *averaged* exact-polar field even though the pointwise map remains singular.

Let

$$L(W) = \frac{1}{2} \|W - C\|_F^2, \quad G(Z, W) = W - C + \sigma Z, \quad (25)$$

where $\text{vec}(Z) \sim N(0, I_{mn})$. Put $R = W - C$. From (10), the frozen input is

$$\mathbf{N}_\beta(W) = R + \sigma \sqrt{c_{\text{Nes}}(\beta)} Z', \quad (26)$$

where Z' is again standard Gaussian and

$$\begin{aligned} c_{\text{Nes}}(\beta) &= (1 - \beta^2)^2 + (1 - \beta)^2 \sum_{j=1}^{\infty} \beta^{2j+2} \\ &= \frac{1 - \beta}{1 + \beta} + 2\beta(1 - \beta)^2. \end{aligned} \quad (27)$$

For ordinary normalized momentum,

$$c_{\text{Mom}}(\beta) = \frac{1 - \beta}{1 + \beta}, \quad c_{\text{Nes}}(\beta) - c_{\text{Mom}}(\beta) = 2\beta(1 - \beta)^2. \quad (28)$$

Thus Nesterov momentum preserves the mean input but increases its frozen variance.

Theorem 5.1 (Gaussian smoothing of the exact-polar field). *Let $\tau = \sigma \sqrt{c_{\text{Nes}}(\beta)} > 0$, let $\mathcal{Q} = \text{Polar}$, and define*

$$\Psi_\tau(R) = \mathbb{E}[\|R + \tau Z\|_*]. \quad (29)$$

Then Ψ_τ is a finite convex C^∞ function and

$$\mathcal{F}_\beta(C + R) = \nabla \Psi_\tau(R) = \mathbb{E}[\text{Polar}(R + \tau Z)]. \quad (30)$$

Moreover, the field is odd and bi-orthogonally equivariant:

$$\begin{aligned} \mathcal{F}_\beta(C - R) &= -\mathcal{F}_\beta(C + R), \\ \mathcal{F}_\beta(C + URV^\top) &= U\mathcal{F}_\beta(C + R)V^\top \end{aligned}$$

for orthogonal U, V of compatible sizes. Along the field ODE for the residual,

$$\dot{R} = -\nabla \Psi_\tau(R), \quad (31)$$

one has

$$\frac{d}{dt} \Psi_\tau(R(t)) = -\|\nabla \Psi_\tau(R(t))\|_F^2, \quad (32)$$

$$\frac{d}{dt} \frac{1}{2} \|R(t)\|_F^2 = -\langle R(t), \nabla \Psi_\tau(R(t)) \rangle_F \leq 0. \quad (33)$$

Proof. The nuclear norm is convex and Lipschitz with respect to the Frobenius norm up to the fixed factor $\sqrt{\min\{m, n\}}$. Convolution with a nondegenerate Gaussian density is finite, convex, and C^∞ . A Gaussian matrix is full rank almost surely, and at a full-rank matrix X the nuclear norm is differentiable with gradient $\text{Polar}(X)$. Differentiation under the expectation, justified by the uniform bound $\|\text{Polar}(X)\|_F \leq \sqrt{\min\{m, n\}}$, proves (30). Symmetry and orthogonal

equivariance follow from the corresponding invariance of the standard Gaussian law. Equation (32) is the chain rule. Finally, convexity implies monotonicity of $\nabla\Psi_\tau$; since symmetry gives $\nabla\Psi_\tau(0) = 0$,

$$\langle R, \nabla\Psi_\tau(R) \rangle_F = \langle R - 0, \nabla\Psi_\tau(R) - \nabla\Psi_\tau(0) \rangle_F \geq 0,$$

which proves (33). $\square$

Theorem 5.1 gives a matrix-level interpretation of noise-induced drift: the field follows the gradient of a smoothed nuclear norm of the residual, not the polar factor of the mean residual. It also supplies a Lyapunov function without assuming that the expected polar direction is pointwise equal to a deterministic steepest-descent direction.

Corollary 5.2 (Closed-form scalar field). *For $m = n = 1$, write $r = w - c$ and let Φ be the standard normal distribution function. Then*

$$\mathcal{F}_\beta(c + r) = 2\Phi(r/\tau) - 1, \quad \dot{r} = -(2\Phi(r/\tau) - 1). \quad (34)$$

The origin is globally attracting, and

$$\mathcal{F}_\beta(c + r) = \sqrt{\frac{2}{\pi}} \frac{r}{\tau} + O(r^3) \quad \text{as } r \rightarrow 0. \quad (35)$$

Hence the additional Nesterov variance in (28) produces a smoother field and a smaller local restoring slope than ordinary momentum.

Proof. In one dimension the polar map is the sign map. Therefore $\mathbb{E}[\text{sign}(r + \tau Z)] = 2\Phi(r/\tau) - 1$. The remaining claims follow from monotonicity and the Taylor expansion of Φ . $\square$

Remark 5.3 (What the Gaussian result does not prove). *The Gaussian gradient in (25) is unbounded, and the pointwise exact polar map is not Lipschitz. Thus Theorem 5.1 identifies and analyzes the field but does not by itself verify the hypotheses of Theorem 4.1. Extending the tracking theorem directly to exact polar under nondegenerate noise requires an averaged regularity argument rather than pointwise Lipschitz propagation.*

6 Discussion and further theoretical problems

Zeros of the field. Even though $\mathbb{E}[\mathbf{N}_\beta(W)] = \nabla L(W)$ for unbiased gradients, generally

$$\mathbb{E}[\mathcal{Q}(\mathbf{N}_\beta(W))] \neq \mathcal{Q}(\nabla L(W)).$$

Consequently, zeros of the matrix field need not coincide with critical points of the objective. This is the same structural phenomenon that makes the Adam vector field different from the mean gradient [8]. The Gaussian quadratic model is a favorable symmetric case in which C is a field zero.

Fixed momentum versus inertial scaling. Theorem 4.1 keeps $\beta < 1$ fixed while $\eta \downarrow 0$, so momentum mixes in $O(1)$ optimizer steps and is averaged out. If $\beta = 1 - \gamma\eta$, the mixing time is $O(\eta^{-1})$ and momentum evolves on the same scale as W . One should then expect a coupled differential system for (W, M) , not the one-variable field ODE.

Exact-polar tracking. There are two plausible routes beyond the present finite-step result. One is to prove tracking for a regularized polar map and pass to zero regularization away from low-rank sets. The other is to exploit smoothing by a nondegenerate innovation law directly at the transition-kernel level. Theorem 5.1 shows that the latter route is not merely formal.

Long-time dynamics and fluctuations. The finite-time law of large numbers is the first step toward an asymptotic pseudo-trajectory theorem, convergence to attracting zeros, and a central limit theorem around the effective flow. The corresponding Adam program is developed in [9, 10]. For Muon, the diffusion coefficient should be the long-run covariance of the stationary sequence $H_w(m_{k-1}^w, U_k)$, including its temporal correlation through momentum.

7 Conclusion

For fixed momentum, stochastic Nesterov Muon is a slow parameter process driven by a fast, geometrically mixing matrix memory. Freezing the slow variable and averaging the nonlinear Nesterov direction produces a deterministic Muon matrix field. Under assumptions that include practical stabilized finite-step Newton–Schulz, the stochastic optimizer tracks the field ODE with an $O(\sqrt{\eta})$ finite-time error. In a Gaussian quadratic model the exact-polar field has an additional variational structure: it is the gradient of a Gaussian-smoothed nuclear norm. These results separate the dynamical question “which field does Muon follow?” from finite-horizon stationarity analysis and provide a rigorous starting point for long-time and fluctuation theory.

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